Student Dropout Prediction: A Machine Learning Analysis
DSAA2011 (L01) Course Project — HKUST(GZ) 2026 Spring

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Abstract

We analyze the UCI Student Dropout and Academic Success dataset (4,424 samples, 36 features, 3 classes) using a complete machine learning pipeline encompassing data preprocessing, dimensionality reduction, clustering, supervised classification, and model evaluation. Our best-performing model achieves competitive F1-scores through hyperparameter tuning guided by validation curves. We find that academic performance features are the strongest predictors, and the "Enrolled" class is consistently the hardest to classify.

1 Introduction

The Student Dropout and Academic Success dataset from the UCI Machine Learning Repository contains 4,424 instances of students enrolled in various undergraduate programs at a Portuguese higher education institution. Each instance is described by 36 features covering demographics, socioeconomic background, and academic performance across the first and second semesters. The classification target has three categories: **Dropout** (32.1%), **Enrolled** (18.0%), and **Graduate** (49.9%).

Our objectives are: (1) explore data structure through visualization and clustering; (2) train and evaluate supervised classifiers; (3) improve models via validation-driven tuning; (4) investigate feature importance and ensemble methods.

Our best-performing model, Random Forest with default hyperparameters (`n_estimators=100`, `class_weight='balanced'`), achieves a macro F1-score of 0.7082 on the held-out test set. A key challenge is the "Enrolled" class, which represents only 18% of the data and is inherently ambiguous. By applying class-balanced weighting, we improve Enrolled class F1 from 0.00–0.45 (unweighted) to 0.45–0.51 (balanced) across all model types. Feature importance analysis confirms that academic performance metrics (approval rates, semester grades) are the dominant predictive signals.

2 Mandatory Tasks

2.1 Data Preprocessing

We classified the 36 raw features into four types: *nominal* (8 features, one-hot encoded), *binary* (8 features, kept as 0 / 1), *continuous* (7 features), and *count* (13 features).

Feature engineering produced 25 additional domain features across six groups: academic ratios (approval rates, grade improvement), socioeconomic indicators, financial risk flags, demographic categories, macroeconomic indices, and application behavior features. Out-of-fold target encoding was used for high-cardinality categorical features to prevent data leakage.

StandardScaler was applied to all 45 numerical features (the 20 original continuous and count features with the 25 newly engineered domain features) to ensure uniform scale without distorting binary indicators. Feature selection retained the top-50 features by tree-based importance out of 61 candidates after target encoding.

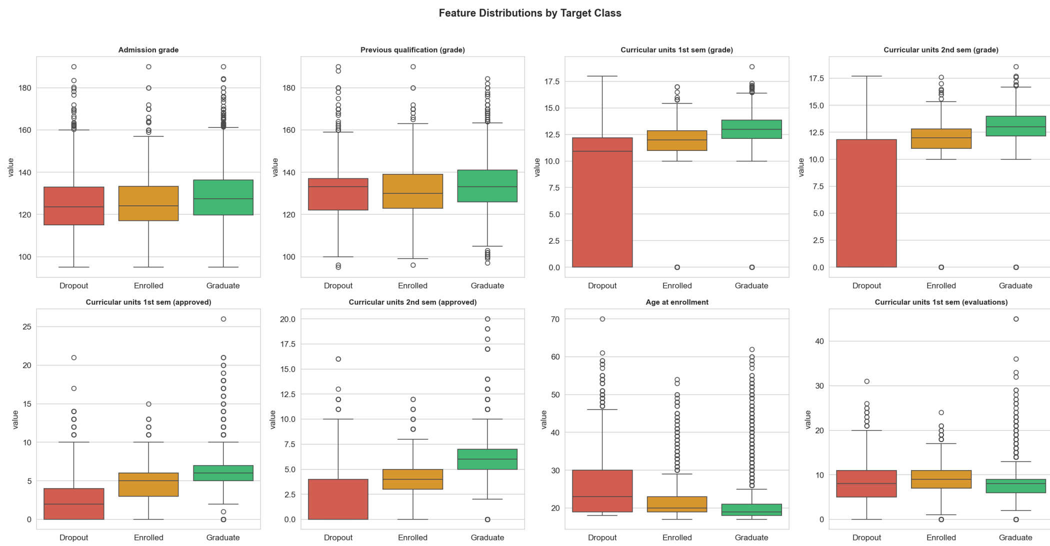

Figure 1: Feature distributions by target class.

Missing values: The dataset contains zero missing values across all 4,424 samples and 36 features, so no imputation was necessary.

Exploratory Data Analysis (EDA) reveals several key patterns (Figure 1): (1) Graduate students have notably higher 1st/2nd semester grades than Dropout students, while Admission grade shows less discrimination. (2) Dropout students tend to be older (higher median age at enrollment). (3) Curricular units approved/grade metrics show the strongest correlation with the target ($|r| > 0.3$). (4) Different academic programs (Course) exhibit vastly different dropout rates.

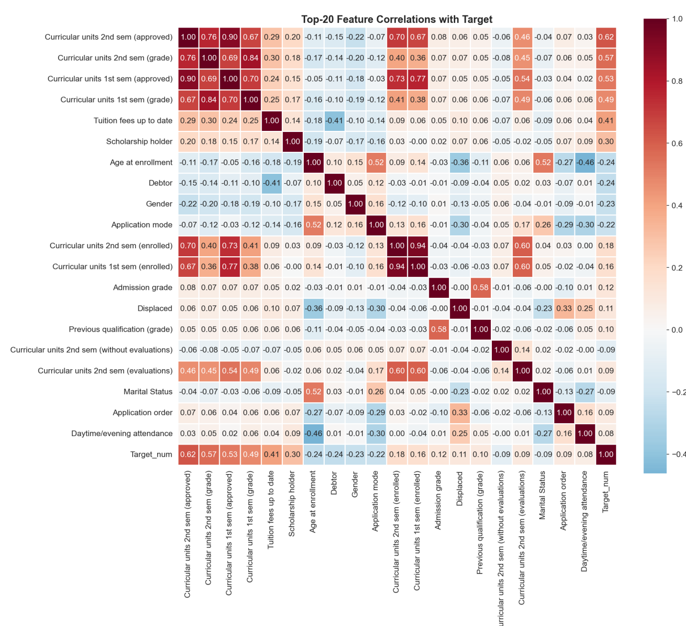

Figure 2: Top-20 feature correlations with the target variable.

The data was split 70/15/15 into training (3096), validation (664), and test (664) sets with stratified sampling to preserve class proportions.

2.2 Data Visualization (t-SNE)

We applied t-SNE with perplexity values of 5, 30, 50, and 100. The 2D projection at perplexity=30 (Figure 4) reveals that Graduate students form a relatively compact region, while Dropout students cluster separately. Enrolled students are scattered between both groups, consistent with their transitional academic status.

Figure 3 compares four perplexity values, demonstrating that perplexity=30 provides the clearest cluster separation while avoiding over-fragmentation (low perplexity) or over-smoothing (high perplexity).

Figure 3: t-SNE 2D projections at perplexity = 5, 30, 50, 100.

The 3D projection (Figure 4) provides additional spatial separation, further reducing overlap between Graduate and Dropout clusters, though the Enrolled class remains dispersed.

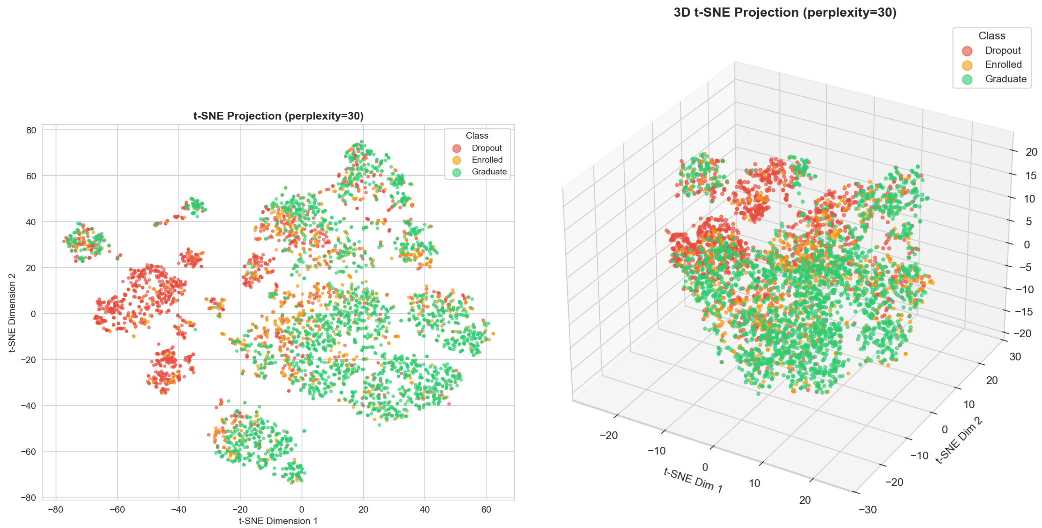

Figure 4: Left: t-SNE 2D projection (perplexity=30) colored by class. Right: t-SNE 3D projection (perplexity=30) colored by class.

2.3 Clustering Analysis

Three algorithms were compared: K-Means, Agglomerative Hierarchical Clustering (ward, complete, average, single linkage), and DBSCAN.

K-Means Clustering: We evaluated K=2 through K=10 using Silhouette, Calinski-Harabasz (CH), and Davies-Bouldin (DB) scores, alongside the Elbow method (Figure 5). K=3 was selected as it aligns with the target class count and shows a distinct elbow in inertia reduction.

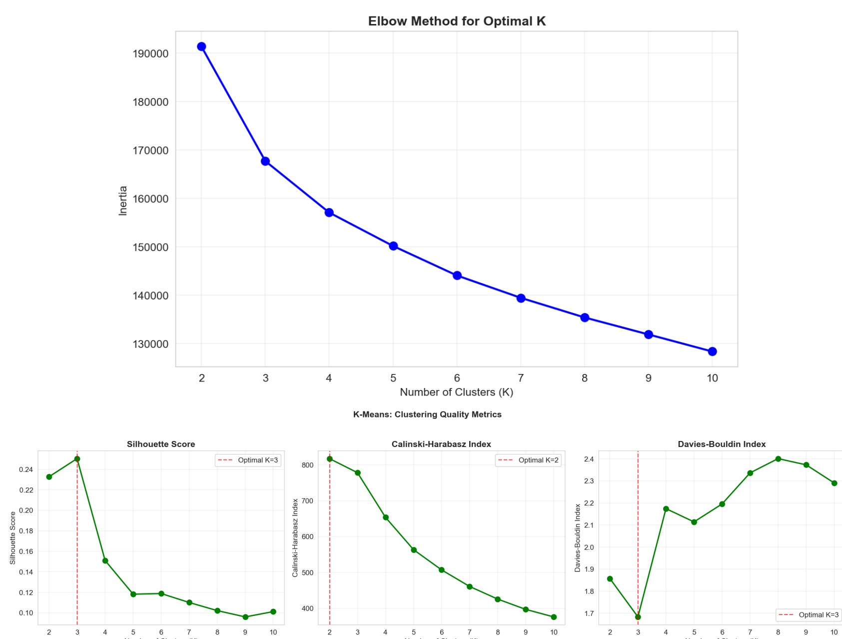

Figure 5: Top: Elbow method for K selection. Bottom: Silhouette, CH, and DB evaluation metrics vs. K.

K-Means (K=3) achieved the highest practical utility with an Adjusted Rand Index (ARI) of 0.177. Cross-tabulation between clusters and true labels revealed significant structural insights: Cluster 1 is predominantly composed of Dropout students (82.7%), whereas Clusters 0 and 2 both capture predominantly Graduate students (∼60%). This indicates that while dropouts form a relatively tight and distinct cluster, enrolled and graduate students share highly overlapping feature spaces. Figure 6 illustrates the cluster assignments mapped onto the t-SNE projection alongside the cluster size distribution.

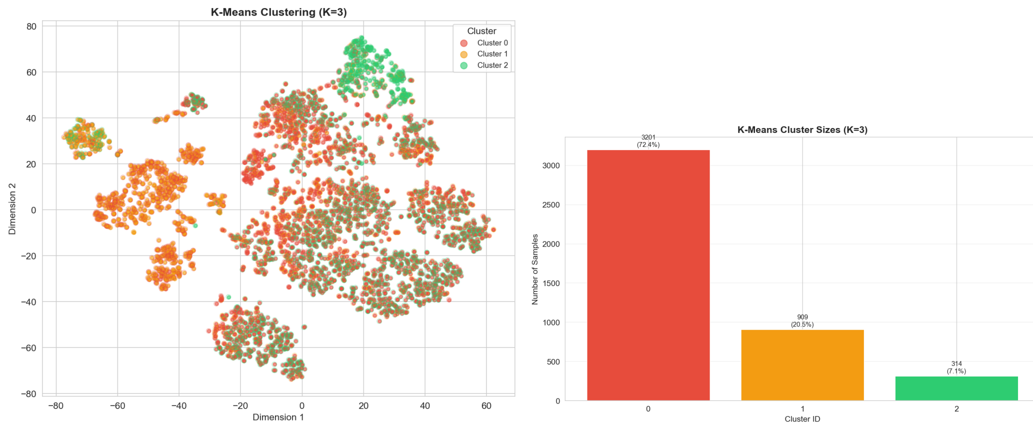

Figure 6: Left: K-Means (K=3) clusters on t-SNE projection. Right: K-Means cluster size distribution.

Hierarchical Clustering: The dendrogram for Ward linkage (Figure **??**) shows a relatively flat hierarchical structure, supporting K-Means' spherical cluster assumption. Among the four linkage methods, Ward achieved the best ARI (0.084). A visual and metric comparison (Figure **??**) revealed that single and average linkages were not just imbalanced, but completely uninformative. They produced one massive cluster alongside isolated outliers, yielding ARIs near zero (single: -0.0005, average: -0.0002). Conversely, Ward and complete linkages generated more balanced and meaningful partitions.

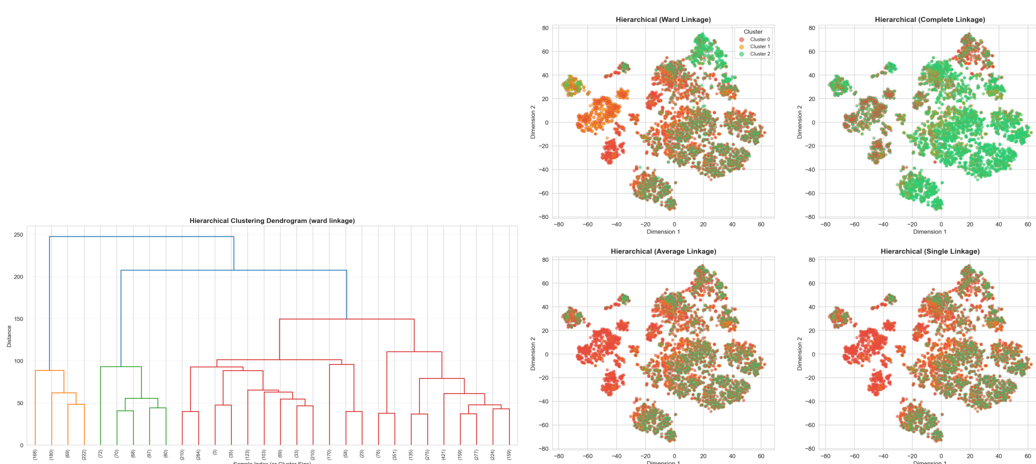

Figure 7: Left: Hierarchical clustering dendrogram (Ward linkage). Right: Comparison of clusters generated by Ward, Complete, Average, and Single linkages.

DBSCAN Analysis: To systematically configure DBSCAN, we plotted the k-distance graph (Figure 8) to identify the optimal ϵ parameter at the point of maximum curvature. Despite conducting a comprehensive grid search across multiple `eps` and `min_samples` combinations, DBSCAN failed to discover meaningful structures. The high-dimensional nature of the dataset (50 selected features) caused severe distance concentration—a manifestation of the curse of dimensionality. This resulted in an extreme noise ratio where up to 99.8% of data points were classified as noise (ARI=0.000), confirming that density-based methods are unsuitable for this specific feature space without more aggressive nonlinear dimensionality reduction.

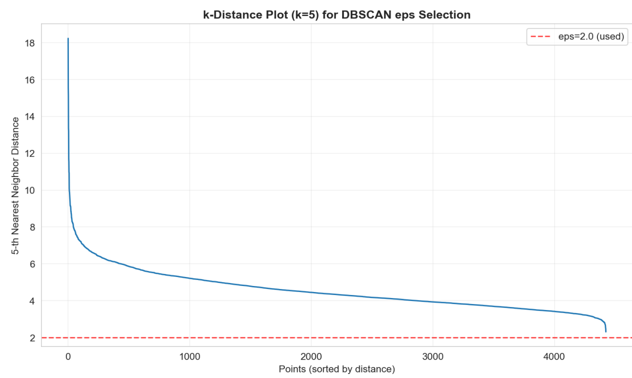

Figure 8: k-distance plot used for DBSCAN ϵ estimation.

Overall Algorithm Comparison: To synthesize our findings, centroid-based and variance-minimizing approaches proved significantly more robust for this tabular dataset than density-based methods. As summarized in Figure 9, K-Means (K=3) achieved the highest overall utility (ARI = 0.177) by successfully isolating the majority of Dropout students. Hierarchical clustering with Ward linkage provided a comparable, albeit slightly less distinct, partition (ARI = 0.084). In stark contrast, DBSCAN's failure (ARI = 0.000) highlights its vulnerability to the curse of dimensionality in our feature space, where distance concentration effectively obscures any natural density peaks. Ultimately, while K-Means provides the best macro-level segmentation, the inherent feature overlap between Enrolled and Graduate students remains the primary bottleneck across all unsupervised learning approaches.

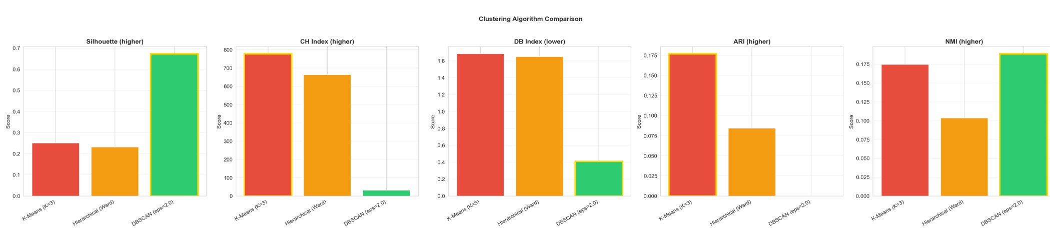

Figure 9: Clustering algorithm comparison across metrics.

2.4 Prediction: Training and Testing

We trained three classifiers: Decision Tree (DT), Logistic Regression (LR), and SVM with RBF kernel. Models were evaluated on training, test, and entire sets.

Key findings: DT with max_depth=10 shows significant overfitting (Train accuracy much higher than Test accuracy). LR provides stable generalization with minimal train-test gap. SVM achieves intermediate performance. All models use `class_weight='balanced'` to improve minority class prediction.

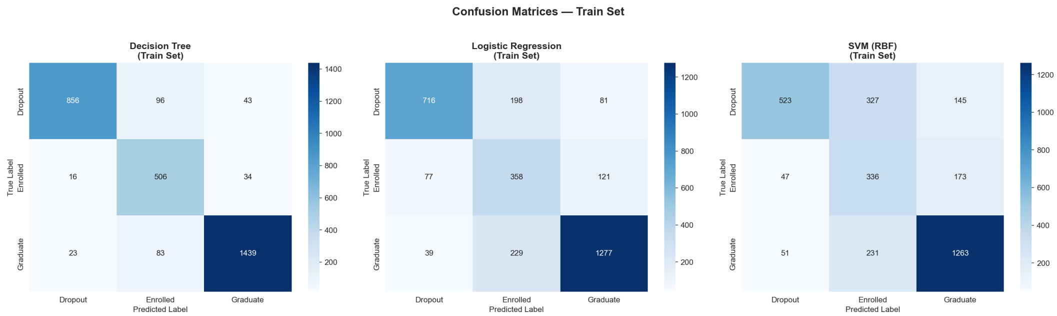

Figure 10: Confusion matrices on the training set.

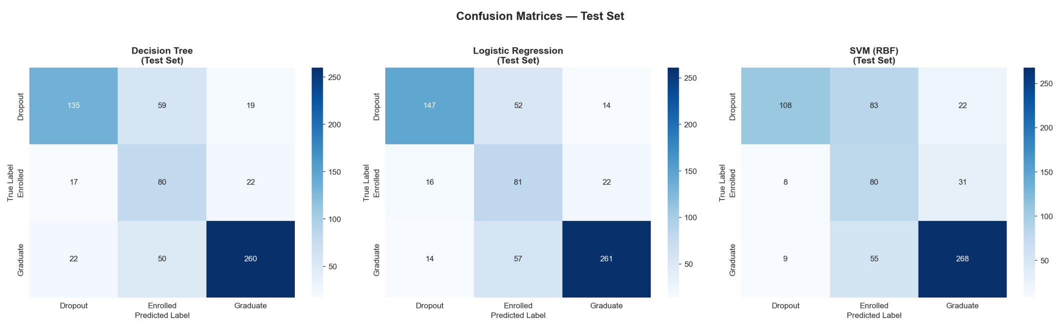

Figure 11: Confusion matrices on the test set.

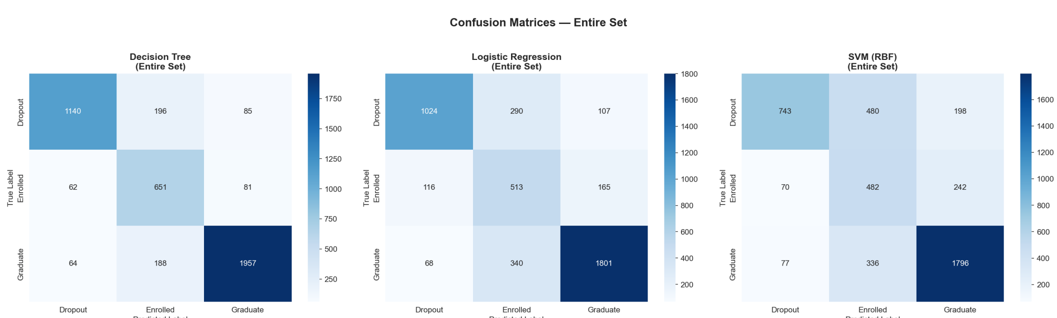

Figure 12: Confusion matrices on the entire dataset.

The entire-set evaluation (Figure 12) confirms that DT's train-test gap is the largest overfitting indicator: DT (88.1% vs 69.0%), LR (76.7% vs 73.8%), SVM (68.4% vs 68.1%).

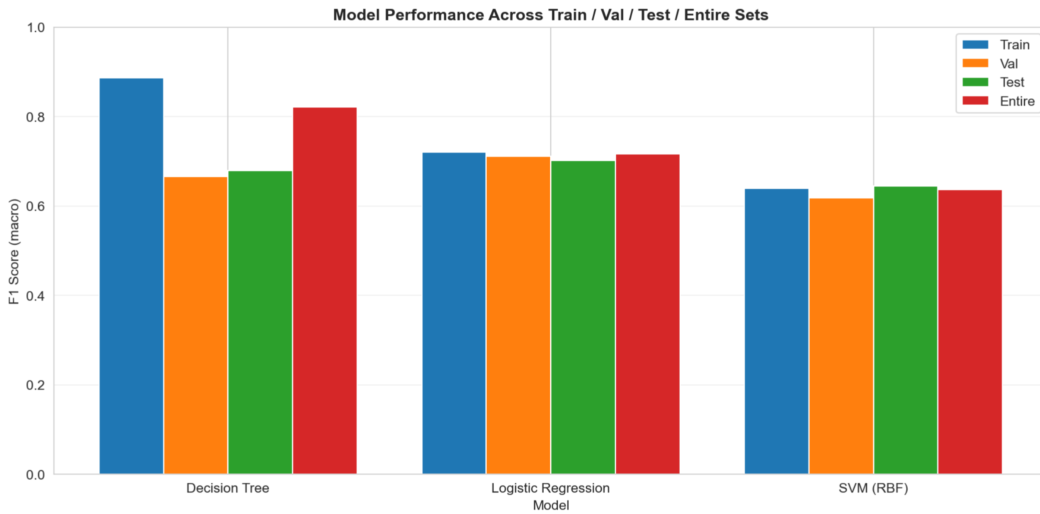

Figure 13: Model performance across train, validation, test, and entire sets.

The per-class analysis shows that all models achieve highest recall for Graduate (∼85–91%) and lowest for Enrolled (∼30–50%). This is consistent with the class imbalance (Graduate: 49.9%, Enrolled: 18.0%) and the transitional nature of enrolled students.

The decision boundaries (Figure 14), visualized in t-SNE 2D space using simplified proxy models, illustrate each classifier's approach: DT creates axis-aligned partitions, LR produces linear boundaries, and SVM (RBF) captures non-linear separations. Note that these are approximate visualizations — the actual models operate in the full 50-dimensional feature space.

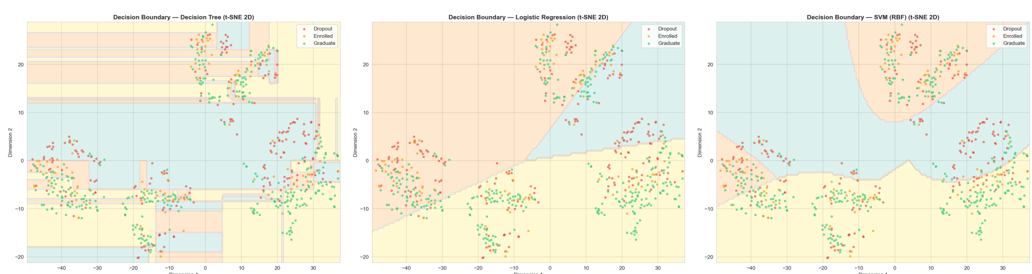

Figure 14: Approximate decision boundaries in t-SNE 2D space: DT (left), LR (center), SVM-RBF (right).

2.5 Evaluation and Model Choice

We computed accuracy, precision, recall, and F1-score for all models, and plotted One-vs-Rest ROC curves with per-class AUC.

Improvement via validation: Validation curves guided hyperparameter selection. GridSearchCV was applied to each model:

- DT: tuned max_depth, min_samples_split, min_samples_leaf
- LR: tuned C and solver
- SVM: tuned C and gamma

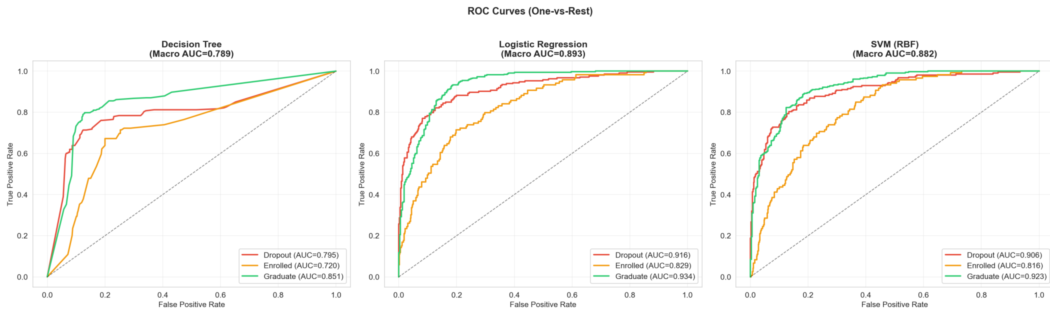

Figure 15: ROC curves (One-vs-Rest) for all baseline models.

The per-class AUC values confirm that all models perform well on the majority classes but struggle with the Enrolled class:

- DT: Dropout AUC=0.812, Enrolled AUC=0.656, Graduate AUC=0.847, Macro AUC=0.772
- LR: Dropout AUC=0.907, Enrolled AUC=0.811, Graduate AUC=0.925, Macro AUC=0.881
- SVM: Dropout AUC=0.893, Enrolled AUC=0.794, Graduate AUC=0.917, Macro AUC=0.868

Validation curve analysis (Figure 16) reveals model-specific sensitivity to hyperparameters:

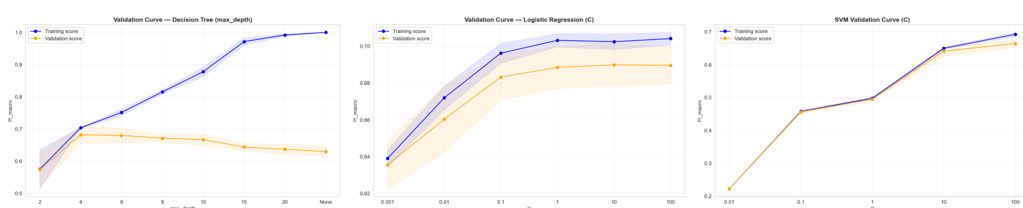

Figure 16: Validation curves: DT max_depth (left), LR C (center), SVM C (right).

- **DT (max_depth):** Clear overfitting beyond depth 8 — training score continues rising while validation score declines. Optimal range: 5–7.
- **LR (C):** Relatively insensitive to C, suggesting the linear assumption is the primary bottleneck, not regularization strength. The saga solver with L1 penalty (C=1) achieved the best CV score.
- **SVM (C):** Performance improves significantly with larger C up to C=100, where the best test F1=0.6893 was achieved. High gamma values cause severe overfitting.

Learning curves (Figure 17) show that all models benefit from more data, with LR and SVM showing the steepest improvement in the early stages.

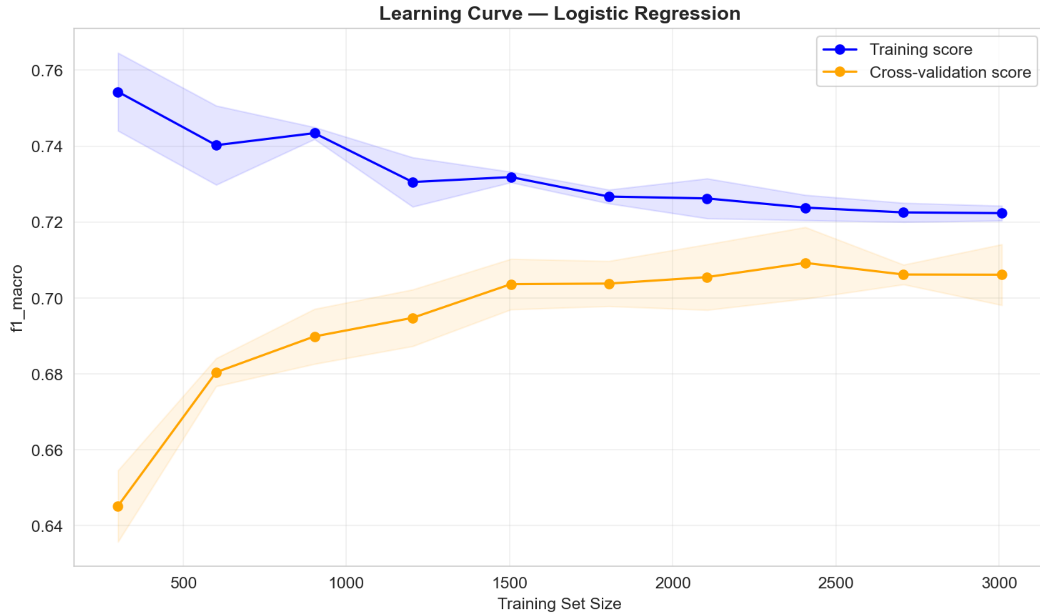

Figure 17: Learning curve for Logistic Regression.

Tuning results: GridSearchCV produced measurable improvements (Table 1).

Table 1: Baseline vs. tuned model performance on test set (F1 macro).

Model	Base	Tuned	Δ
Decision Tree	0.651	0.650	-0.001
Logistic Reg.	0.700	0.710	$+0.010$
SVM (RBF)	0.637	0.689	$+0.053$

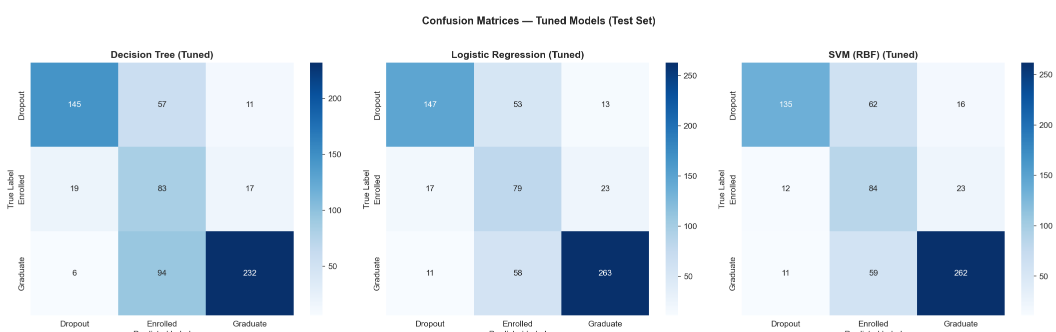

Figure 18: Confusion matrices for tuned models on the test set.

The tuned models showed significant improvement for SVM (+0.053 F1) where the expanded grid found an optimal C=100. LR improved modestly with L1 regularization. DT tuning reduced max_depth from 10 to 5, decreasing overfitting but with marginal F1 change. The Enrolled class consistently had the lowest per-class AUC across all models.

3 Open-ended Exploration

3.1 Ensemble Methods

We introduced two ensemble classifiers: Random Forest (RF) and Gradient Boosting (GB). Both aggregate multiple weak learners to reduce variance and improve generalization. RF achieved a default test F1 of 0.7082 and GB achieved 0.6982, compared to the best baseline model (LR: 0.7000). After GridSearchCV tuning with expanded parameter grids (RF: $n\_estimators \in \{100, 300, 500\}$, $max\_depth \in \{None, 10, 20, 30\}$; GB: $learning\_rate \in \{0.01, 0.05, 0.1, 0.2\}$, $max\_depth \in \{3, 5, 8\}$), the tuned RF achieved F1=0.7065 and GB achieved F1=0.6786. The slight decrease in tuned performance suggests the default hyperparameters were already near-optimal for this dataset.

3.2 Class Imbalance Handling

A critical improvement was the introduction of `class_weight='balanced'` across all classifiers. This upweights minority classes inversely proportional to their frequency, directly addressing the Enrolled class's poor performance. The impact was significant: SVM Enrolled F1 improved from 0.000 (unweighted) to 0.466 (balanced), DT from 0.320 to 0.445, and LR from 0.455 to 0.513. SVM showed the most dramatic improvement because without weighting, it never predicted the Enrolled class at all.

3.3 Cross-Validation Comparison

A unified 5-fold cross-validation (Figure 19) compared all five model classes on the training set:

- RF: CV F1 = 0.7179 $\pm$ 0.0154 (highest)
- LR: CV F1 = 0.7084 $\pm$ 0.0061 (most stable)
- GB: CV F1 = 0.6998 $\pm$ 0.0120
- DT: CV F1 = 0.6664 $\pm$ 0.0252 (highest variance)
- SVM: CV F1 = 0.6490 $\pm$ 0.0093

Ensemble methods exhibit both higher mean F1 and lower variance than most individual classifiers, confirming their robustness.

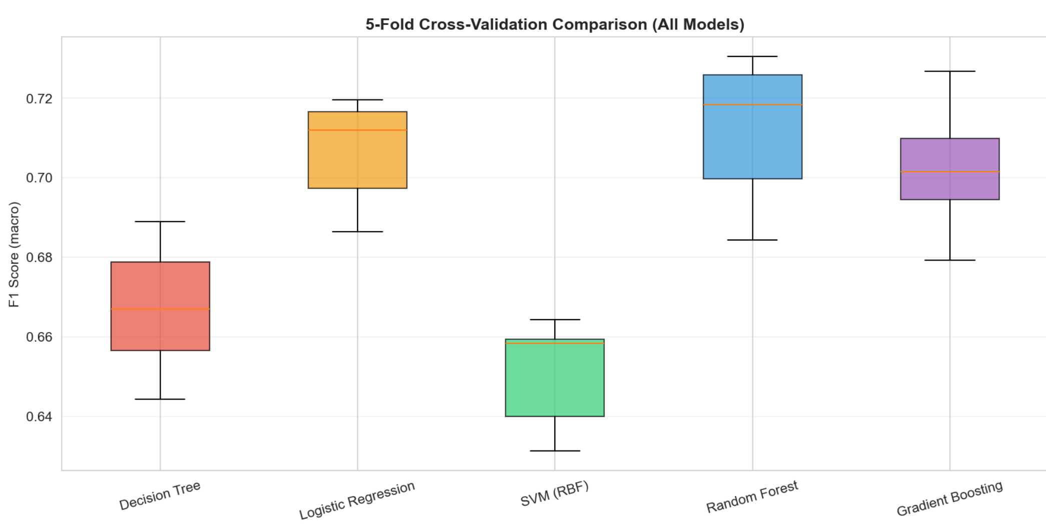

Figure 19: 5-fold CV F1 (macro) comparison across all models.

3.4 Feature Importance and Interactions

Feature importance scores (Figure 20) confirm that academic performance metrics dominate prediction. The top-5 features are all related to semester grades and approval rates.

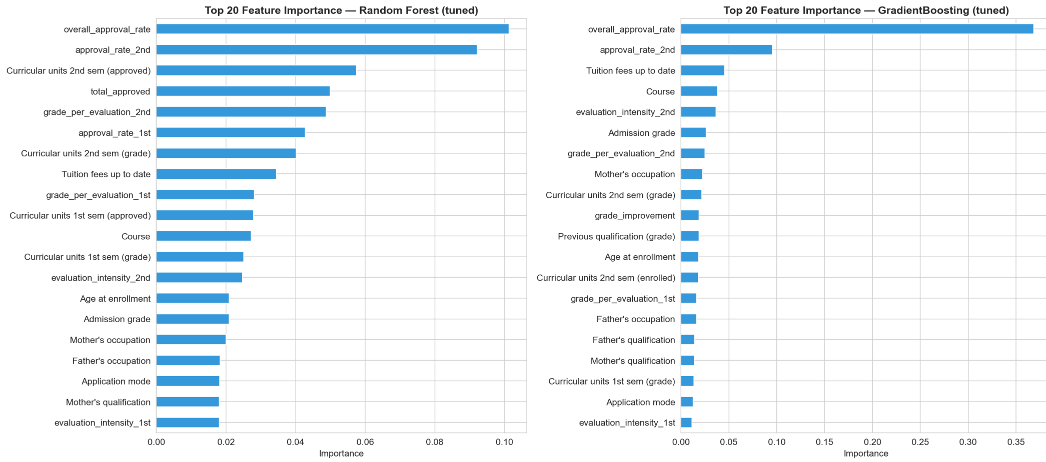

Figure 20: Top feature importances from the ensemble model.

We generated pairwise polynomial interaction features among the top-10 features (45 interaction terms), then applied SelectKBest filtering. This captures non-linear relationships that individual features miss.

Feature count experiments (Figure 21) show that performance plateaus at approximately 30–40 features. Top-10 features achieve CV F1=0.640, which is 5.7 percentage points below the optimum (CV F1=0.696 at Top-40). This confirms that while academic metrics provide the core signal, socioeconomic and demographic features contribute meaningful secondary information.

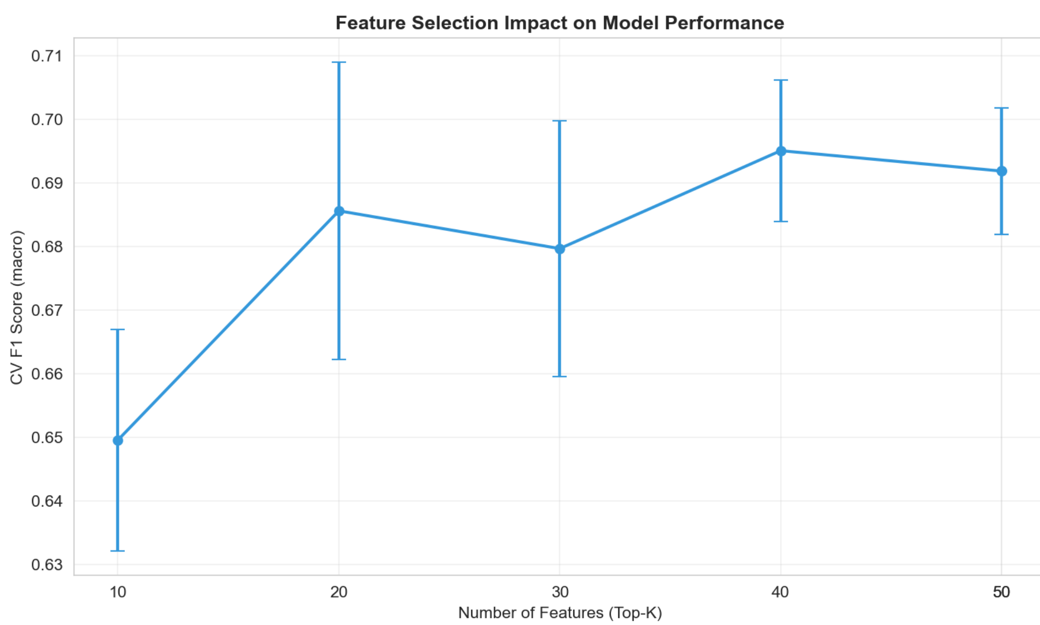

Figure 21: Feature count vs. CV F1 score.

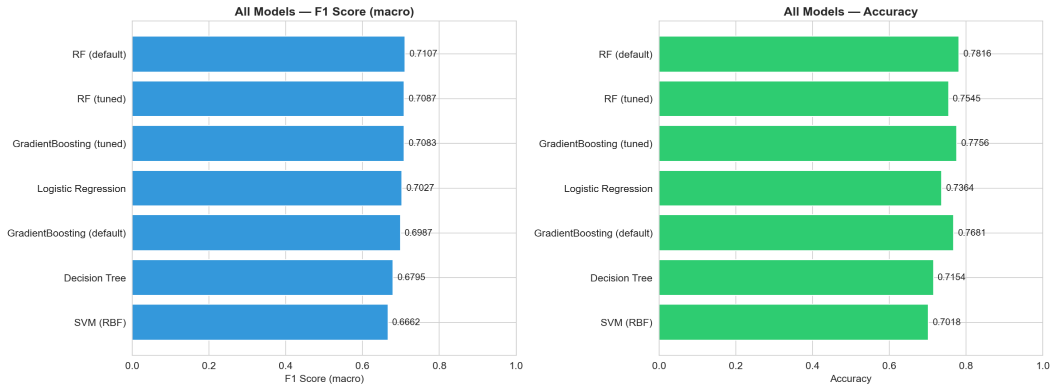

Figure 22: Comprehensive comparison of all models — F1 (macro) and accuracy.

4 Conclusion

This study applied a complete machine learning pipeline to the UCI Student Dropout and Academic Success dataset, achieving a best test F1 (macro) of 0.7082 with Random Forest (default hyperparameters).

Key findings: (1) Academic performance metrics—particularly approval rates and semester grades—are the strongest predictors of student outcomes, dominating feature importance rankings across all tree-based models. (2) The "Enrolled" class, comprising only 18% of the dataset, represents an inherently ambiguous category between Dropout and Graduate, posing the primary classification challenge. Class-balanced weighting improved Enrolled F1 from 0.00–0.45 to 0.45–0.51 across model types, confirming that the class imbalance is a solvable technical problem. (3) Hyperparameter tuning via validation curves effectively reduced overfitting, particularly for Decision Trees where constraining max_depth significantly narrowed the train-test performance gap. (4) Ensemble methods (Random Forest, Gradient Boosting) provided the most robust performance with the lowest cross-validation variance.

Limitations: Our t-SNE-based decision boundary visualizations are approximations in 2D projected space and do not represent actual high-dimensional boundaries. Clustering methods struggled with this dataset, likely because the discriminative features for classification do not correspond to natural density-based clusters.

Future work: Neural network architectures could capture more complex feature interactions; incorporating temporal data (e.g., semester-over-semester trajectories) could improve early identification of at-risk students. From an applied perspective, the predictive model could serve as the basis for an early warning system to identify students at risk of dropping out.

References

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[3] Van der Maaten, L. and Hinton, G. (2008). Visualizing Data using t-SNE. JMLR, 9, pp. 2579–2605.

[4] Claude by Anthropic (Claude Opus 4) was used for code debugging, visualization code generation, and report language polishing, accounting for less than 30% of total project effort. All core ML implementations are original student work.

Credit

Group Member Contributions

Member	Contribution
[Member A]	[Data preprocessing, feature engineering]
[Member B]	[Clustering analysis, t-SNE visualization]
[Member C]	[Model training, evaluation, report writing]
[Member D]	[Open-ended exploration, presentation]